

Module 1: Cyber Security Introduction

Section 1: Core Concepts

1. What is the meaning of cyber security?

Cyber security is the practice of protecting systems, networks, programs, and data from digital attacks. These "cyberattacks" are usually aimed at accessing, changing, or destroying sensitive information, extorting money from users, or interrupting normal business processes.

2. What are the main objectives of cyber security?

The main objectives are governed by the **CIA Triad**:

- **Confidentiality:** Ensuring that data is accessible only to authorized individuals.
- **Integrity:** Ensuring that information is accurate, complete, and has not been tampered with.
- **Availability:** Ensuring that systems and data are accessible to users when needed.

3. What is offensive and defensive in cyber security?

- **Offensive Security (Red Teaming):** Proactive measures that involve ethical hacking to find vulnerabilities before a real attacker does. It involves "attacking" one's own systems to test strength.
- **Defensive Security (Blue Teaming):** Reactive and proactive measures focused on protecting the infrastructure. This includes firewalls, patching, monitoring, and incident response.

4. What is cyberspace and law?

- **Cyberspace:** The global, virtual environment of interconnected computers, networks, and data where communication and social interaction occur.
- **Cyber Law:** The legal framework that deals with the internet, computing, and related legal issues, such as privacy, freedom of expression, and intellectual property.

5. What is cyber warfare?

Cyber warfare involves the use of digital attacks by one nation-state to disrupt the computer systems of another, with the intent of causing significant damage, financial loss, or intelligence theft.

Section 2: Hackers & Operations

6. Explain the Types of Hacker.

- **White Hat:** Ethical hackers who use their skills for good, helping organizations find and fix security holes (with permission).
- **Black Hat:** Malicious hackers who break into systems for personal gain, data theft, or destruction.
- **Grey Hat:** Hackers who may break the law to find a vulnerability but don't have malicious intent; they often report the bug to the owner for a fee.

7. What is the full form of SOC in cyber security?

Security Operations Center. It is a centralized unit that deals with security issues on an organizational and technical level.

8. What are the Challenges of Cyber Security?

- **Evolving Threats:** Ransomware and AI-driven attacks are constantly changing.
- **Data Proliferation:** The massive amount of data being generated makes it harder to protect.
- **Skill Gap:** A shortage of trained cyber security professionals.
- **Remote Work:** Securing home networks and personal devices used for work.
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Module 2: Fundamentals of OS & Networks

Section 3: Hardware & Networking

1. Difference between hardware and software.

- **Hardware:** The physical components of a computer system (e.g., CPU, RAM, Hard Drive, Monitor). You can touch it.
- **Software:** A set of instructions or programs that tell the hardware what to do (e.g., Windows, Chrome, Python scripts). It is virtual.

2. Define IP address range and private address range.

- **IP Address Range:** The full span of addresses available (IPv4 ranges from 0.0.0.0 to 255.255.255.255).
- **Private Address Range:** Specific ranges reserved for internal local area networks (LANs) that are not routable on the public internet.

- Class A: \$10.0.0.0\$ – \$10.255.255.255\$
- Class B: \$172.16.0.0\$ – \$172.31.255.255\$
- Class C: \$192.168.0.0\$ – \$192.168.255.255\$

3. Explain Network protocol and Port number.

- **Network Protocol:** A set of rules that governs how data is exchanged between devices (e.g., TCP, IP, UDP).
- **Port Number:** A logical address used to identify a specific process or service on a device (e.g., Port 80 for Web, Port 22 for SSH).

4. Explain Types of Network Devices.

- **Router:** Connects different networks (like your home network to the internet).
- **Switch:** Connects devices within the same network.
- **Hub:** A basic device that broadcasts data to all connected ports (rarely used now).
- **Firewall:** Monitors and filters incoming/outgoing traffic based on security rules.

5. Which Tools use for Data Backup and Recovery?

- **Backup Tools:** EaseUS Todo Backup, Acronis Cyber Protect, Veeam.
- **Recovery Tools:** Recuva, TestDisk, Stellar Data Recovery.

Section 4: Security Protocols

6. Explain HTTP and HTTPS Protocols.

- **HTTP (Hypertext Transfer Protocol):** The standard protocol for transferring web pages. Data is sent in plaintext (unsafe).
- **HTTPS (HTTP Secure):** The secure version of HTTP. It encrypts the communication using SSL/TLS to prevent eavesdropping.

7. What is SSL and TLS Security?

- **SSL (Secure Sockets Layer):** The original encryption protocol for the web. It is now considered deprecated (outdated).
- **TLS (Transport Layer Security):** The modern, more secure successor to SSL. It provides data encryption, integrity, and authentication for internet communication.

8. Explain the MAC ADDRESS.

Media Access Control (MAC) Address. It is a unique, permanent physical address assigned to a Network Interface Controller (NIC) by the manufacturer. It consists of 12 hexadecimal characters (e.g., 00:1A:2B:3C:4D:5E).

